

Physics-informed neural electron closures for multiscale heliospheric modelling

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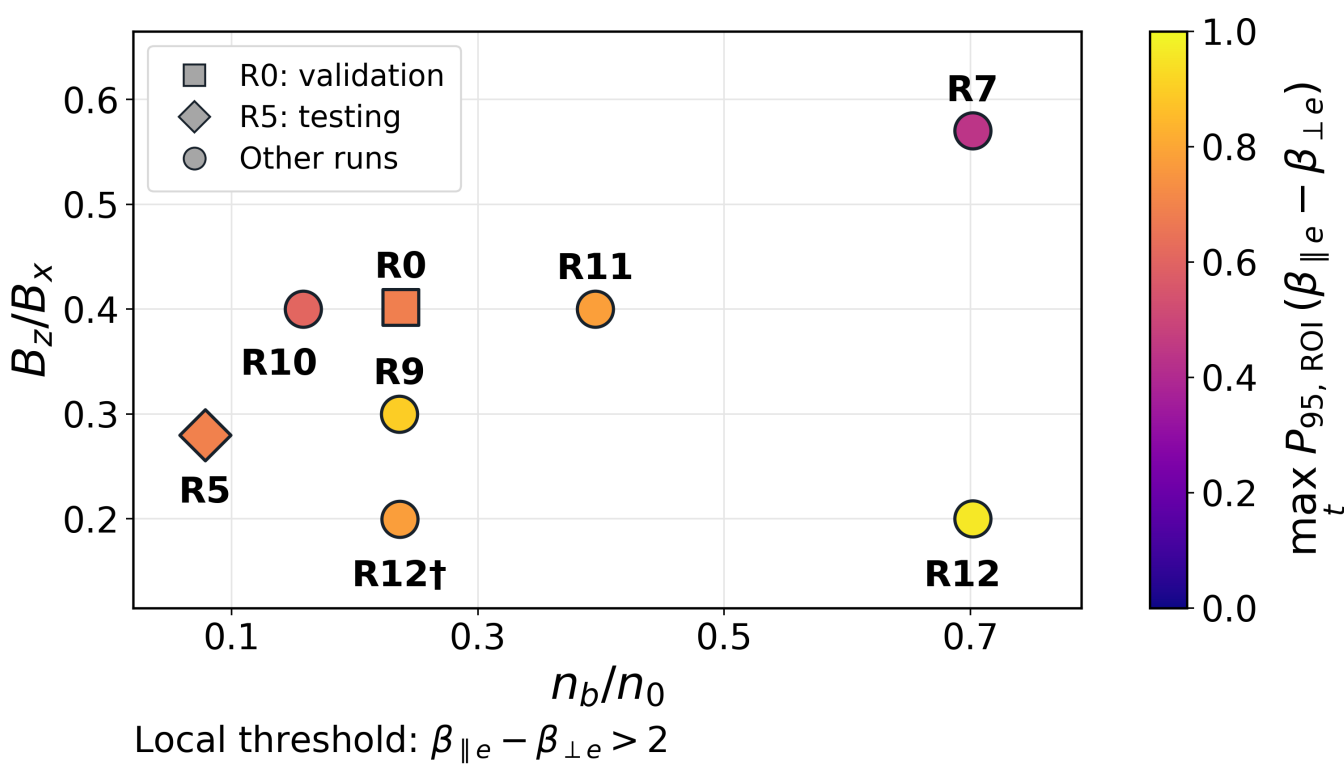
Abstract

Processes in the heliosphere, such as magnetic reconnection, are fundamentally multiscale. In particular, electron pressure anisotropies result in feedback that drives out-of-plane current. Traditional fluid and hybrid kinetic frameworks with isothermal electrons cannot capture such feedback. We address this challenge by performing a family of fully kinetic semi-implicit PIC simulations of 2D magnetic reconnection and use machine learning to extract a phenomenological equation of state [1] for the electron pressure tensor and test it “online” in hybrid PIC model Menura. We also introduce spectral stability analysis based on auto-differentiation.

Fully Kinetic run of Harris sheet [2]

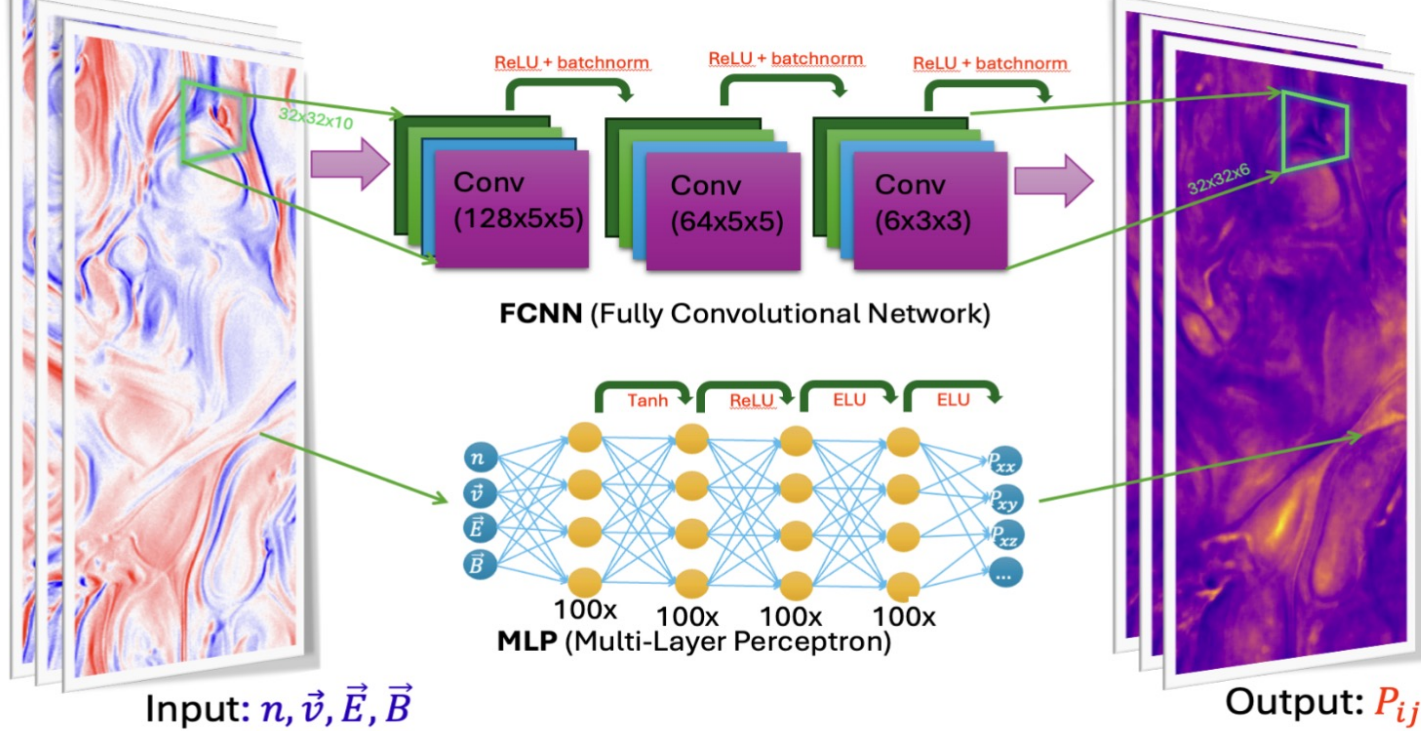
We run parameter sweep for Harris sheet reconnection over guide field B_z and background density n_b using Energy conserving PIC [2]

Electron firehose criterion



Pressure Tensor Learned [3]

We learn pressure tensor as a function of lower order moments



Autodiff for spectral radius ρ

● **Automatic differentiation**: linear response of pressure
$$\delta \mathbf{P}_e = \frac{\partial \mathbf{P}_\theta}{\partial \mathbf{B}} \bigg|_{\text{background}} \delta \mathbf{B}. \quad (1)$$

● **Solver map $\mathbf{G}$** : Follow the disturbance through one solver macrostep
$$\delta \hat{\mathbf{B}}^{m+1} = \mathbf{G}_T(\mathbf{k}) \delta \hat{\mathbf{B}}^m. \quad (2)$$

● **Spectral radius**: maximum eigenvalue of the solver map
$$\rho_{\max} = \max_{\text{tested backgrounds, } k, j} |\lambda_j(\mathbf{G}_T)|. \quad (3)$$

Conclusions

● We confirm with semi-implicit fully kinetic code ECsim [2] that in moderate guide-field reconnection pressure anisotropy at firehose condition drives out-of-plane current not consistent with isothermal hybrid PIC run

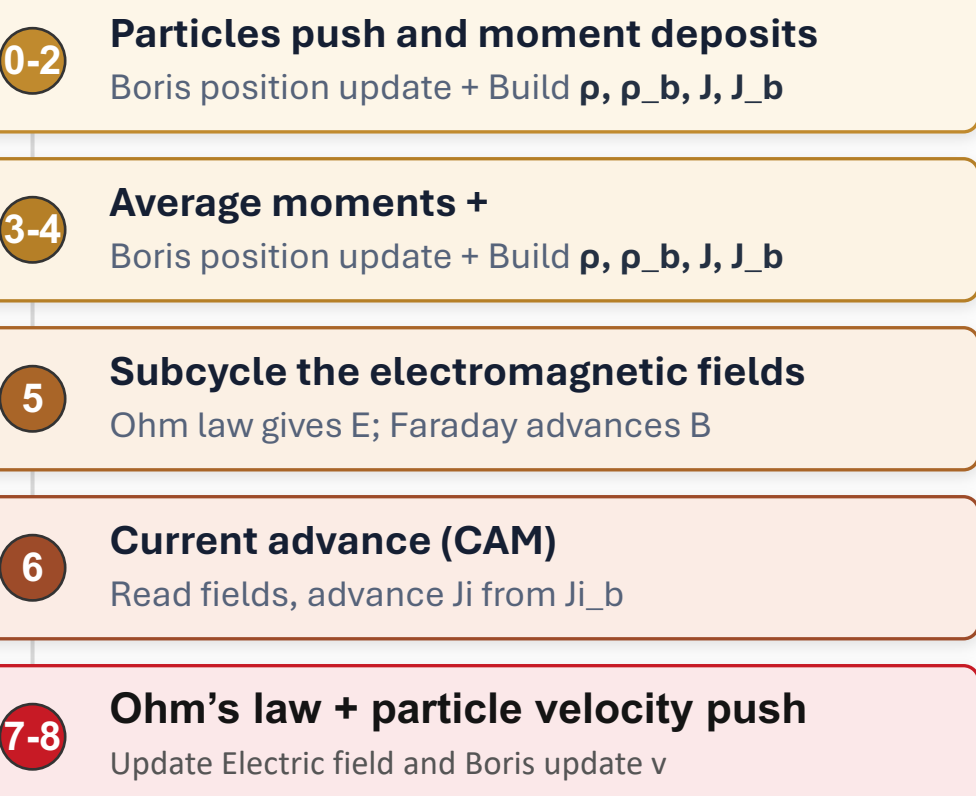
● We show that it is possible to obtain qualitatively consistent driving in hybrid PIC code when supplying neural closure trained on fully kinetic simulations, but local model (MLP) is the stable one despite higher loss

● New autodiff-based linear response theory for neural closure derives criterion for stability which qualitatively confirms that MLP outperforms global model FCNN online and that shallower FCNN should blow-up.

Online implementation of electron neural closure in Menura

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E} \quad \mathbf{E} = -\mathbf{u}_i \times \mathbf{B} + \frac{\mathbf{J}_{\text{tot}} \times \mathbf{B}}{en} - \frac{\nabla \cdot \mathbf{P}_e}{en} + \eta \mathbf{J}_{\text{tot}} - \eta_h \nabla^2 \mathbf{J}_{\text{tot}} \quad (4)$$

Full macrostep of hybrid PIC model



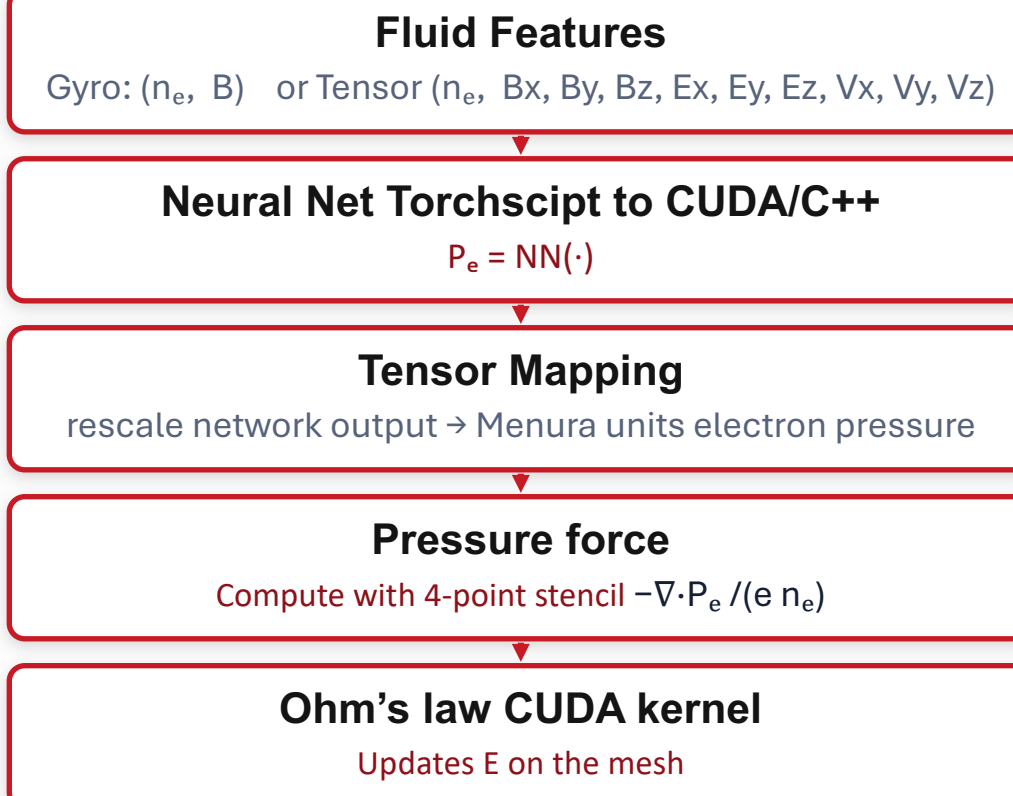
Closure Call #1

Closure Call #2

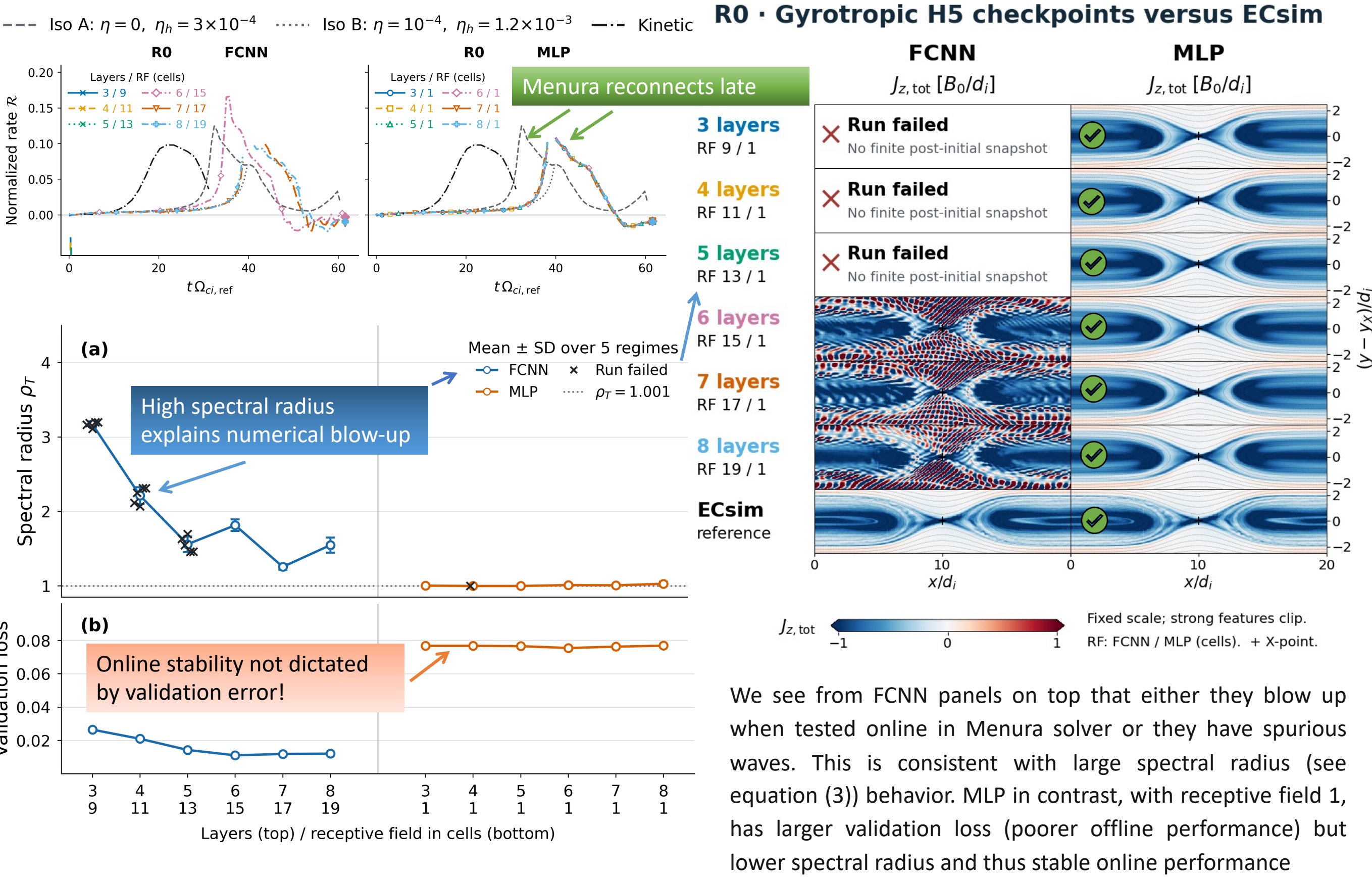
Hold pressure
Saves compute during subcycling

Closure Call #3

Learned neural NN electron closure



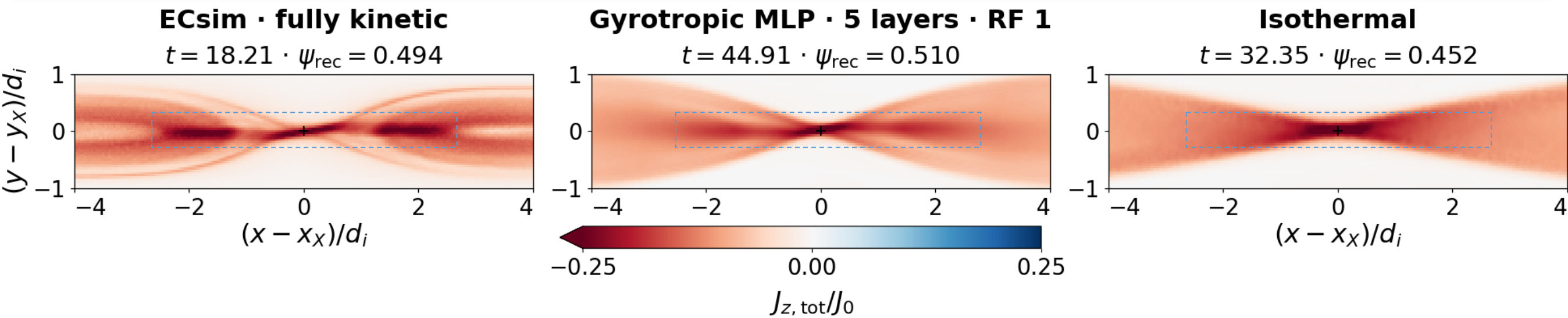
For spectral linear response theory we autodifferentiate Menura macrostep assuming frozen ions. We conclude that gyrotropic pressure model (GYRO) is significantly more stable $\mathbf{P}_e = p_\perp \mathbf{I} + (p_\parallel - p_\perp) \hat{\mathbf{b}} \hat{\mathbf{b}}^T$, $(p_\parallel, p_\perp) = \text{NN}_\theta(n, |\mathbf{B}|)$, $\hat{\mathbf{b}} = \frac{\mathbf{B}}{|\mathbf{B}|}$. (5)



We see from FCNN panels on top that either they blow up when tested online in Menura solver or they have spurious waves. This is consistent with large spectral radius (see equation (3)) behavior. MLP in contrast, with receptive field 1, has larger validation loss (poorer offline performance) but lower spectral radius and thus stable online performance

R5: Out-of-plane current feedback with Gyrotropic MLP closure

$$\mathbf{J}_{\perp,e} \simeq -ne \frac{\mathbf{E} \times \mathbf{B}}{B^2} + \frac{\mathbf{B}}{B^2} \times \nabla p_{\perp,e} + \frac{\mathbf{B}}{B^2} \times (p_{\parallel,e} - p_{\perp,e}) (\mathbf{b} \cdot \nabla) \mathbf{b} + \frac{\mathbf{B}}{B^2} \times \mathbf{b} \mathbf{b} \cdot \nabla (p_{\parallel,e} - p_{\perp,e}) + \frac{\mathbf{B}}{B^2} \times (\nabla \cdot \mathbf{\Pi}_e) \quad (6)$$



REFERENCES

- [1] Le, A., et al. Physics of Plasmas **23**, 2016.
- [2] Lapenta, G., et al J Comp Phys. **334**, 2017
- [3] Miloshevich, G., et al. Physics of Plasmas **33**, 2026

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Kinetic runs performed on the VSC Tier-1 system Hortense.

georgemilosh.github.io

